

18/12/2022

MODULE-03

LAPLACE'S AND POISSON'S EQUATION.

Derivation of Laplace's and Poisson's equation.

From point form of Gauss law,

$$\nabla \cdot \mathbf{D} = \rho_v \longrightarrow (1)$$

Since $\mathbf{D} = \epsilon_0 \mathbf{E}$ and $\mathbf{E} = -\nabla V$, the expression (1) can be re-written as,

$$\nabla \cdot (\epsilon_0 \mathbf{E}) = \rho_v$$

$$\nabla \cdot [\epsilon_0 (-\nabla V)] = \rho_v$$

$$\nabla \cdot \nabla V = \frac{-\rho_v}{\epsilon_0}$$

$$\boxed{\nabla^2 V = \frac{-\rho_v}{\epsilon_0}} \longrightarrow (2)$$

This is called the Poisson's equation.

The double ∇ operation is expanded in rectangular co-ordinate system as,

$$\nabla \cdot \mathbf{A} = \frac{\partial (A_x)}{\partial x} + \frac{\partial (A_y)}{\partial y} + \frac{\partial (A_z)}{\partial z}$$

$$\nabla V = \frac{\partial (V)}{\partial x} \mathbf{a}_x + \frac{\partial (V)}{\partial y} \mathbf{a}_y + \frac{\partial (V)}{\partial z} \mathbf{a}_z$$

$$\therefore \boxed{\nabla \cdot \nabla V = \frac{\partial}{\partial x} \left(\frac{\partial V}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial V}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial V}{\partial z} \right)} \longrightarrow (3)$$

The expression (3) can be re-written as,

$$\nabla \cdot \nabla V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = \frac{-\rho_v}{\epsilon_0} \longrightarrow (4)$$

If $S_r = 0$, the expression ④ becomes,

$$\boxed{\nabla^2 V = 0}$$

This is called the Laplace equation.

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• The Laplace equation in all 3 Co-ordinate System

1. Rectangular Co-ordinate System.

In RCS,

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0.$$

2. Cylindrical Co-ordinate System.

In CCS,

$$\nabla^2 V = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) + \frac{1}{s^2} \left(\frac{\partial^2 V}{\partial \phi^2} \right) + \frac{\partial^2 V}{\partial z^2}$$

3. Spherical Co-ordinate System.

In SCS,

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}.$$

• Problems:

1. Verify whether the potential function, $V = 2x^2 - 4y^2 + z^2$ and $V = r \cos \phi + z$. Satisfies Laplace equation or not.

Q1 $V = 2x^2 - 4y^2 + z^2$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\frac{\partial(V)}{\partial x} = 4x + 0 + 0 = \underline{4x}$$

$$\frac{\partial^2 V}{\partial x^2} = \underline{4}$$

$$\frac{\partial(V)}{\partial y} = \underline{-8y}$$

$$\frac{\partial^2 V}{\partial y^2} = \underline{-8}$$

$$\frac{\partial(V)}{\partial z} = \underline{2z}$$

$$\frac{\partial^2 V}{\partial z^2} = \underline{2}$$

$$\therefore \nabla^2 V = 4 - 8 + 2$$

$$\nabla^2 V = -2 \neq 0$$

$\therefore$ The given potential function doesn't satisfy Laplace equation.

$$V = r \cos \phi + z$$

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2 V}{\partial \phi^2}$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \left(\frac{\partial^2 V}{\partial \theta^2} \right) + \frac{\partial^2 V}{\partial \phi^2}$$

$$\frac{\partial V}{\partial r} = 0$$

$$\frac{\partial V}{\partial \theta} = r (-\sin \theta) \Rightarrow \frac{\partial^2 V}{\partial \theta^2} = -r (\cos \theta)$$

$$\frac{\partial v}{\partial z} = 1 \Rightarrow \frac{\partial^2 v}{\partial z^2} = 0$$

$$\nabla^2 v = 0 + \frac{1}{s^2} (-s \cos \theta) \neq 0$$

$$\boxed{\nabla^2 v = -\frac{s \cos \theta}{s^2} \neq 0}$$

∴ This doesn't satisfy Laplace equation

2) $v = r^2 \cos \phi + \theta$

$$\nabla^2 v = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v}{\partial \phi^2}$$

$$\frac{\partial v}{\partial r} = \cos \phi (2r) \Rightarrow \frac{\partial}{\partial r} (r^2 \cos \phi \cdot 2r) = \frac{\partial}{\partial r} (2r^3 \cos \phi) = 2 \cos \phi (3r^2) = 6 \cos \phi r^2$$

$$\frac{\partial v}{\partial \theta} = 1$$

$$\frac{\partial v}{\partial \phi} = r^2 (-\sin \phi) \Rightarrow \frac{\partial^2 v}{\partial \phi^2} = -r^2 \cos \phi$$

$$\begin{aligned} \nabla^2 v &= \frac{1}{r^2} (6 \cos \phi r^2) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta) + \frac{1}{r^2 \sin^2 \theta} (-r^2 \cos \phi) \\ &= 6 \cos \phi + \frac{1}{r^2 \sin \theta} (\cos \theta) + \frac{\cos \phi}{\sin^2 \theta} \end{aligned}$$

$$\boxed{\nabla^2 v = 6 \cos \phi + \frac{\cos \theta}{r^2 \sin \theta} + \frac{\cos \phi}{\sin^2 \theta}}$$

5) Find the potential and the volume charge density at $P(0.5, 1.5, 1)$ mtr. in free space. The given potential is $V = 68\phi^2$ volts.

sol $P(0.5, 1.5, 1)$ mtr.

$$V = 68\phi^2$$

$$S = \sqrt{x^2 + y^2} = \sqrt{0.5^2 + 1.5^2} = \underline{\underline{\sqrt{2.5} \text{ mtr.}}}$$

$$\phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \left(\frac{1.5}{0.5} \right) = \underline{\underline{71.56^\circ}}$$

$$z = z = \underline{\underline{1 \text{ mtr}}}$$

$$P(0.5, 1.5, 1) \longleftrightarrow P(\sqrt{2.5}, 71.56^\circ, 1)$$

$$P(0.5, 1.5, 1) \longleftrightarrow P(\sqrt{2.5}, 1.249, 1)$$

$$V \text{ at } P(\sqrt{2.5}, 1.249, 1)$$

$$\therefore V = 6 \times \sqrt{2.5} \times 1.249 \times 1 = \frac{(468) \times 0.2}{10}$$

$$\boxed{V = 11.84 \text{ V}}$$

$$\nabla^2 V = 0$$

$$\nabla^2 V = -\frac{\rho_v}{\epsilon_0}$$

$$\rho_v = -\nabla^2 V \epsilon_0$$

$$\nabla^2 V \text{ in } \text{ECS,}$$

$$\rho_v = - \left[\frac{1}{S} \frac{\partial}{\partial S} \left(S \left(\frac{\partial V}{\partial S} \right) \right) + \frac{1}{S^2} \left(\frac{\partial^2 V}{\partial \phi^2} \right) + \frac{\partial^2 V}{\partial z^2} \right] \epsilon_0$$

$$= - \left[\frac{1}{S} \frac{\partial}{\partial S} (8.6\phi^2) + \frac{1}{S^2} (0) + 0 \right] \epsilon_0$$

$$= \left[\frac{1}{3} (6\phi z) + 0 + 0 \right] \epsilon_0$$

$$S_v = \left(-\frac{6\phi z}{3} \right) \epsilon_0$$

$$S_v = -\frac{6\phi z}{3} (8.854 \times 10^{-12})$$

$$S_v = \frac{(-6)(1.24)(1)}{\sqrt{2.5}} (8.854 \times 10^{-12})$$

$$\underline{S_v = -41.96 \times 10^{-2} \text{ V/m}}.$$

A) Show $\nabla^2 v = 0$ for $v = (5s^4 - 6s^{-4}) \sin 4\phi$

Sol $\nabla^2 v$ in CCS

$$\nabla^2 v = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial v}{\partial s} \right) + \frac{1}{s^2} \left(\frac{\partial^2 v}{\partial \phi^2} \right) + \cancel{\frac{\partial^2 v}{\partial z^2}}$$

$$\frac{\partial v}{\partial s} = (20s^3 + 24s^{-5}) \sin 4\phi$$

$$s \frac{\partial}{\partial s} \left(s \frac{\partial v}{\partial s} \right) = (20s^4 + 24s^{-4}) \sin 4\phi$$

$$\frac{\partial}{\partial s} \left(s \frac{\partial v}{\partial s} \right) = \underline{(80s^3 - 96s^{-5}) \sin 4\phi}$$

$$\frac{\partial v}{\partial \phi} = (5s^4 - 6s^{-4}) (4 \cos 4\phi)$$

$$\frac{\partial^2 v}{\partial \phi^2} = (5s^4 - 6s^{-4}) (-16 \sin 4\phi).$$

$$\therefore \nabla^2 v = \frac{1}{s} (80s^3 - 96s^{-5}) \sin 4\phi + \frac{1}{s^2} (5s^4 - 6s^{-4}) (-16 \sin 4\phi)$$

$$= 80s^2 \sin 4\phi - 96s^{-6} \sin 4\phi + \cancel{80s^2 \sin 4\phi} + 96s^{-6} \sin 4\phi$$

$$\boxed{\nabla^2 v = 0} \quad \therefore \underline{\text{It Satisfies Laplace equation.}}$$

5. Calculate Numerical Values for V and S at point P in free space if $V = \frac{4yz}{x^2+1}$ at $P(1, 2, 3)$.

$$V = \frac{4yz}{x^2+1} \text{ at } P(1, 2, 3)$$

$$V_p = \frac{4(2)(3)}{(1)^2+1} = \underline{12V}$$

$$\therefore \boxed{V_p = 12V}$$

$$\nabla^2 V = \frac{\rho}{\epsilon_0}$$

$$S = -\nabla^2 V \epsilon_0$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x} [4yz(x^2+1)^{-1}]$$

$$= 4yz [-1(x^2+1)^{-2} (2x+0)]$$

$$\frac{\partial V}{\partial x} = -8xyz(x^2+1)^{-2}$$

$$\frac{\partial^2 V}{\partial x^2} = -8xyz [-2(x^2+1)^{-3} (2x+0)]$$

$$= +32x^2yz(x^2+1)^{-3}$$

$$\boxed{\frac{\partial^2 V}{\partial x^2} = \frac{32x^2yz}{(x^2+1)^3}}$$

$$\frac{\partial^2 V}{\partial y^2} = -8xz [x(-2(x^2+1)^{-3}) + (x^2+1)^{-2} (1)]$$

$$= -8xz (-2x(x^2+1)^{-3}) + (x^2+1)^{-2}$$

$$\boxed{\frac{\partial^2 V}{\partial x^2} = +32x^2yz(x^2+1)^{-3} + 8yz(x^2+1)^{-2}}$$

$$\boxed{\frac{\partial^2 V}{\partial y^2} = 0}$$

$$\boxed{\frac{\partial^2 V}{\partial z^2} = 0}$$

$$\rho = -\nabla^2 V \epsilon_0$$

$$= - (32x^2y z (x^2+1)^{-3} - 8yz(x^2+1)^{-2}) 8.854 \times 10^{-12}$$

$$\text{at } x=1, y=2, z=3$$

$$\rho = -106.24 \text{ pC/m}^3$$

6. Calculate numerical value of ρ and V in free space at

i) $V = 5 \rho^2 \cos(2\phi)$ at $P(3, \frac{\pi}{3}, 2)$

ii) $V = 2 \cos \phi$ at $P(0.5, 45^\circ, 60^\circ)$

Sol i) ρ, ϕ, z
 $P(3, \frac{\pi}{3}, 2)$

$$V = 5 \rho^2 \cos(2\phi)$$

$$= 5 (3)^2 \cos(2 \times \frac{\pi}{3})$$

$$V = -22.5 \text{ V}$$

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \left(\frac{\partial^2 V}{\partial \phi^2} \right) + \frac{\partial^2 V}{\partial z^2}$$

$$\frac{\partial V}{\partial \rho} = \frac{\partial}{\partial \rho} 5 \rho^2 \cos(2\phi)$$

$$\frac{\partial V}{\partial \rho} = 10 \cos(2\phi) (\rho)$$

$$\frac{\partial V}{\partial \phi} = 5 \rho^2 (-2 \sin 2\phi)$$

$$= -10 \rho^2 \sin 2\phi$$

$$\frac{\partial^2 V}{\partial \phi^2} = -10 \rho^2 (2 \cos 2\phi)$$

$$\frac{\partial^2 V}{\partial \phi^2} = -20 \rho^2 \cos 2\phi$$

$$\frac{\partial^2 V}{\partial z^2} = 0$$

$$\nabla^2 V = \frac{1}{s} \frac{\partial}{\partial s} \left[10 s^2 \cos(2\phi) \right] + \frac{1}{s^2} (-20 s^2 \cos 2\phi) + 0$$

$$= \frac{1}{s} [10 \cos 2\phi (2s)] - 20 \cos 2\phi$$

$$= 20 \cos 2\phi - 20 \cos 2\phi$$

$$\nabla^2 V = 0$$

$$\therefore [s=0]$$

i) $V = \frac{2 \cos \phi}{s^2}$ at $P(0.5, 45^\circ, 60^\circ)$ deg $\rightarrow$ rad.
 $P = (0.5, 0.78, 1.04)$ (Degree $\times \frac{\pi}{180}$)

$$V = \frac{2 \cos(1.04)}{(0.5)^2}$$

$$V = 4.04 \text{ V}$$

$$\nabla^2 V = \frac{1}{s^2} \frac{\partial}{\partial s} \left(s^2 \frac{\partial V}{\partial s} \right) + \frac{1}{s^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{s^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

$$\frac{\partial V}{\partial s} = \frac{\partial}{\partial s} (2 \cos \phi s^{-2}) = 2 \cos \phi (-2s^{-3})$$

$$\frac{\partial V}{\partial s} = -4 s^{-3} \cos \phi$$

$$s^2 \frac{\partial V}{\partial s} = -4 s^{-1} \cos \phi$$

$$\frac{\partial}{\partial s} \left(s^2 \frac{\partial V}{\partial s} \right) = +4 s^{-2} \cos \phi$$

$$\frac{\partial}{\partial s} \left(s^2 \frac{\partial V}{\partial s} \right) = \frac{4 \cos \phi}{s^2}$$

$$\frac{\partial v}{\partial \theta} = 2 \cos \phi r^{-2}$$

$$\boxed{\frac{\partial^2 v}{\partial \theta^2} = 0}$$

$$\frac{\partial v}{\partial \phi} = 2 \cos \phi r^{-2}$$

$$= 2(r^{-2})(-\sin \phi)$$

$$\frac{\partial^2 v}{\partial \phi^2} = 2(r^{-2})(-\cos \phi)$$

$$\frac{\partial^2 v}{\partial \phi^2} = \frac{-2 \cos \phi}{r^2}$$

$$\therefore \nabla^2 v = \frac{1}{r^2} \left(\frac{4 \cos \phi}{r^2} \right) + 0 + \frac{1}{r^2 \sin^2 \theta} \left(\frac{-2 \cos \phi}{r^2} \right)$$

$$= \frac{4 \cos \phi}{r^4} - \frac{2 \cos \phi}{r^4 \sin^2 \theta}$$

$$\nabla^2 v = \frac{2 \cos \phi}{r^4} \left[2 - \frac{1}{\sin^2 \theta} \right]$$

$$\rho = (-\nabla^2 v) \epsilon_0$$

$$= - \left[\frac{2 \cos \phi}{r^4} \left(2 - \frac{1}{\sin^2 \theta} \right) \right] \times 8.854 \times 10^{-12}$$

$$\text{at } r = 0.5, \theta = \frac{\pi}{4}, \phi = \frac{\pi}{3}$$

$$\boxed{\rho = -6.26 \text{ pC/m}^3}$$

$$\boxed{\rho = 0}$$

7. For a given field potential $V = 2x^2 + 5y^2$ at the point $P(5, 1, 3)$. Find the numerical values of ρ and S .

Q1 $V = 2x^2 + 5y^2$
 $P = (5, 1, 3)$

$$V = 2(5)^2 + 5(1)^2$$

$$V = 55 \text{ V}$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\frac{\partial V}{\partial x} = 4x$$

$$\frac{\partial V}{\partial y} = 10y$$

$$\frac{\partial^2 V}{\partial z^2} = 0$$

$$\frac{\partial^2 V}{\partial x^2} = 4$$

$$\frac{\partial^2 V}{\partial y^2} = 10$$

$$\nabla^2 V = 4 + 10 + 0$$

$$\nabla^2 V = 14$$

$$\rho = -\nabla^2 V \times \epsilon_0$$

$$\rho = -(14) \cdot (8.854 \times 10^{-12})$$

$$\rho = -123.956 \text{ pC/m}^3$$

• APPLICATIONS OF LAPLACE EQUATION:

1) Solve Laplace equation for potential field in the homogeneous region between the two concentric conducting spheres with radii a & b such that $b > a$. If the potential $V = 0$ at $r = b$ and $V = V_0$ at $r = a$ also find the capacitance between two concentric spheres.

(or)

2) find the capacitance b/w two concentric sphere of radii, $r = b$, $r = a$, such that $b > a$ if the potential $V = 0$ at $r = b$, using the Laplace equation.

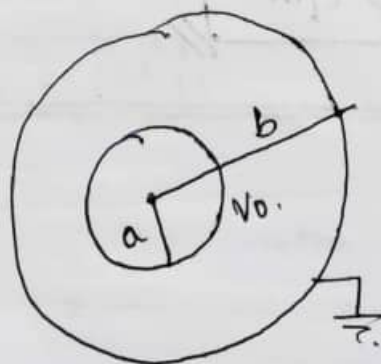
(or)

3) By applying Laplace equation, find the expression for capacitance between 2 concentric sphere, make suitable assumptions.

(or)

4) Solve the Laplace equation for the potential field and find the capacitance in homogeneous region between two concentric conducting sphere with radii a & b . Such that $b > a$, if $V = 0$ at $r = b$ and $V = V_0$ at $r = a$.

Sol



$a < b$ $b > a$

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial V}{\partial r} \right] + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta \frac{\partial V}{\partial \theta} \right] + \frac{1}{r^2 \sin \theta} \frac{\partial^2 V}{\partial \phi^2} = 0$$

Since V is the function of r the second and third terms in the above expression are zero.

$$\therefore \nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial V}{\partial r} \right] \rightarrow \textcircled{1}$$

Since r is not equal to zero ($r \neq 0$), $r^2 \neq 0$, $\frac{1}{r^2} \neq 0$

Therefore the expression 1 can be written as

$$\frac{\partial}{\partial r} \left[r^2 \frac{\partial V}{\partial r} \right] = 0 \rightarrow \textcircled{2}$$

Integrate (2) w.r.t r

$$\int \frac{\partial}{\partial r} \left[r^2 \frac{\partial V}{\partial r} \right] = \int 0 \rightarrow \textcircled{2}$$

$$r^2 \frac{\partial V}{\partial r} = C_1$$

$$\frac{\partial V}{\partial r} = \frac{C_1}{r^2} \rightarrow \textcircled{3}$$

Integrate (3) w.r.t r again

$$V = -\frac{C_1}{r} + C_2 \rightarrow \textcircled{4}$$

Using the Boundary Conditions that is,

$r = a$ $V = -\frac{C_1}{a} + C_2$ $V_0 = -\frac{C_1}{a} + C_2$ $\textcircled{5a}$		$r = b$ $V = -\frac{C_1}{b} + C_2$ $0 = -\frac{C_1}{b} + C_2$ $\textcircled{5b}$
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Subtract $0 - V_0$.

$$0 - V_0 = -\frac{C_1}{b} + \cancel{0} + \frac{C_1}{a} - \cancel{C_2}$$

$$-V_0 = \frac{C_1}{a} - \frac{C_1}{b}$$

$$V_0 = +C_1 \left[+\frac{1}{b} - \frac{1}{a} \right] \rightarrow \textcircled{6}$$

$$C_2 = \frac{C_1}{b} \rightarrow \textcircled{7}$$

from $\textcircled{6}$,

$$C_1 = \frac{V_0 (ab)}{a-b} \rightarrow \textcircled{8}$$

from $\textcircled{7}$

$$C_2 = \frac{V_0 (a)}{a-b} \rightarrow \textcircled{9}$$

Substitute $\textcircled{8}$ and $\textcircled{9}$ in $\textcircled{4}$
 $\therefore$ The potential V is given by,

$$V = -\frac{C_1}{x} + C_2$$

$$= -\frac{V_0(ab)}{(a-b)x} + \frac{V_0(a)}{(a-b)}$$

$$= -\frac{V_0(ab)}{x(a-b)} + \frac{V_0(a)}{(a-b)}$$

$$x = a$$

$$V = \frac{-V_0(ab)}{a(a-b)} + \frac{V_0(a)}{(a-b)}$$

$$C = \frac{Q}{V}$$

$$C = \frac{Q}{\frac{-V_0(ab)}{a(a-b)} + \frac{V_0(a)}{(a-b)}}$$

$$= \frac{Q}{\frac{-V_0(ab)(a-b) + V_0(a) a(a-b)}{a(a-b)^2}}$$

$$= \frac{Q(a(a-b)^2)}{-V_0(ab)(a-b) + V_0(a^2)(a-b)}$$

$$C = \frac{Q a(a-b)}{-V_0(ab) + V_0(a^2)}$$

Apply

2. To Calculate 'C' of parallel plate Capacitor.

Q) Two parallel plate Capacitor Separated by a distance d mtr. and maintained at a potential 0 and V_0 respectively assuming negligible fringing Capacitance effect determine potential at any point between the plates and the Capacitance.

(or)

Q) Apply Laplace equation to

obtain the expression for Capacitance of parallel plate Capacitor, the distance between the plates are d and area of plates is A

$$\nabla^2 V = 0$$

↓
Find V

↓
Apply Boundary condition

↓
Find C_1, C_2

↓
Find V B/w 2 plates

↓
Find E

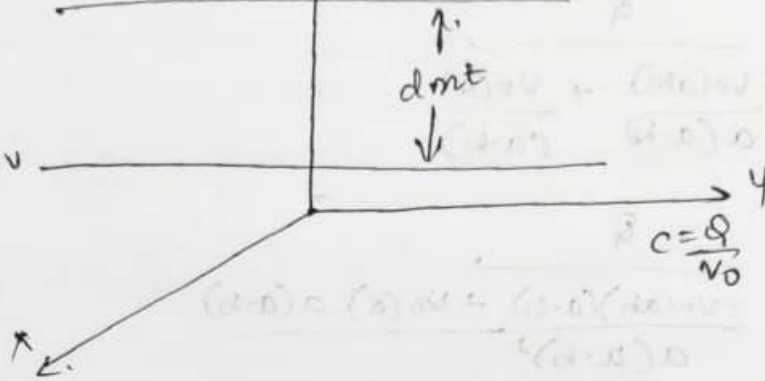
↓
Find Q

↓
Find C

Ans

$$z = d \text{ mtd, } V = \frac{V_0 V}{2}$$

$$z = 0, V = 0 \text{ V}$$



$$\nabla^2 V = \frac{\partial^2 V}{\partial z^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial x^2} = 0$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial z^2} = 0 \rightarrow (1)$$

$$\int \frac{\partial^2 V}{\partial z^2} = \int 0$$

$$\frac{\partial V}{\partial z} = C_1 \rightarrow (2)$$

$$V = C_1 z + C_2 \rightarrow (3)$$

at $z = 0, V = 0 \text{ V}$

$$V = C_1 z + C_2$$

$$0 = C_1(0) + C_2$$

$$\boxed{C_2 = 0} \rightarrow (4)$$

at $z = d, V = V_0$

$$V_0 = C_1(d) + C_2$$

$$V_0 = C_1(d) + 0$$

$$\boxed{C_1 = \frac{V_0}{d}} \rightarrow (5)$$

Substituting C_1 & C_2 in eqn (3)

$$\Rightarrow V = \frac{V_0}{d}(z) + 0$$

$$\boxed{V = \frac{V_0 z}{d}} \rightarrow (6)$$

Using the expression 6, Voltage Between any point between two plates,

$$E = -\nabla V = -\frac{\partial V}{\partial z} \hat{a}_z$$

from (2),

$$E = -C_1 \hat{a}_z \rightarrow (7)$$

from (5),

$$\boxed{E = -\frac{V_0}{d} \hat{a}_z \text{ V/m}} \rightarrow (8)$$

Since $D = \epsilon_0 E = \frac{Q}{A}$

$$|D| = \epsilon_0 |E| = \frac{Q}{A}$$

$$\epsilon_0 \left(\frac{V_0}{d} \right) = \frac{Q}{A}$$

$$\therefore \boxed{Q = \frac{\epsilon_0 V_0 A}{d}} \rightarrow (9)$$

$$C = \frac{Q}{V_0}$$

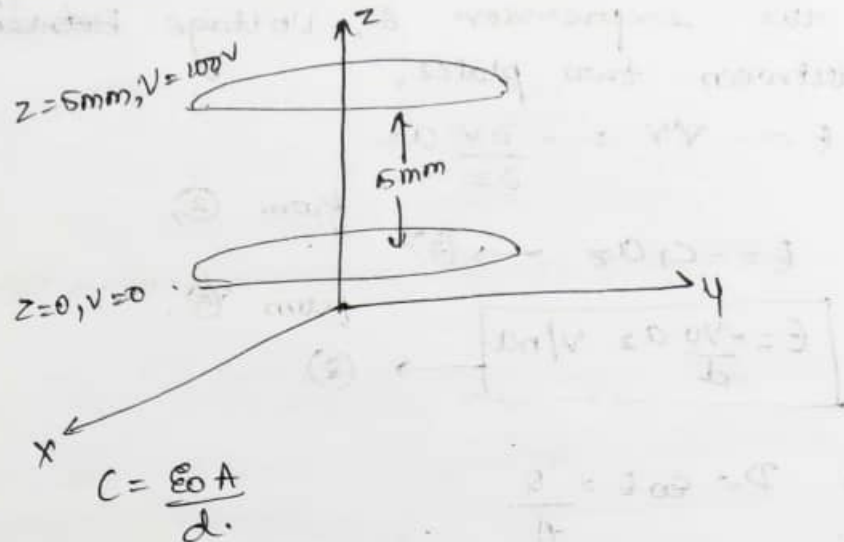
$$= \frac{\epsilon_0 V_0 A}{d V_0}$$

$$\boxed{C = \frac{\epsilon_0 A}{d}} \rightarrow (10)$$

→ Problems

1. Two parallel conducting discs are separated by a distance 5mm at $z=0$ and $z=5\text{mm}$ by a distance 5mm at $z=0$ and $z=5\text{mm}$. $V_0=0$ and $V=100\text{V}$ respectively. And it is only in z -direction. Starting from Laplace equation find the capacitance and the surface charge density ρ_s on the discs.

Q1



$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial z^2}$$

$$\int \frac{\partial^2 V}{\partial z^2} = \int 0$$

$$V = C_1 z + C_2$$

at $V = 0, z = 0$

$$0 = C_1(0) + C_2$$

$$\boxed{C_2 = 0}$$

at $z = 5\text{mm}, V = 100\text{V}$

$$100 = C_1(5\text{mm}) + C_2$$

$$C_1 = \frac{100}{5 \times 10^{-3}}$$

$$\boxed{C_1 = 20\text{K}}$$

$$V = 20\text{K}(z) + 0$$

$$\boxed{V = 20\text{K}z}$$

$$E = -\nabla V = -\frac{\partial V}{\partial z} a_z$$

$$E = -C_1 a_z$$

$$E = -20k a_z \text{ V/m}$$

$$D = \epsilon_0 E = \frac{Q}{A}$$

$$\therefore \epsilon_0 \times 20k = \frac{Q}{A}$$

$$Q = \epsilon_0 20k A$$

$$C = \frac{Q}{V_0}$$

$$C = \frac{\epsilon_0 20k A}{V_0}$$

$$V_0 = (100 - 0) = 100$$

$$C = \frac{\epsilon_0 20k A}{100} = \frac{200 \epsilon_0 A}{100}$$

$$C = 1.77 \text{ nF}$$

$$S_0 = \epsilon_0 E$$

$$|S_0| = \epsilon_0 |E|$$

$$= \epsilon_0 (20k)$$

$$|S_0| = 1.77 \times 10^{-8} \text{ W/m}^2$$

3. To find the Capacitance of Co-axial Cable

a) Using the Laplace equation, derive an expression for capacitance per unit length of Co-axial cable. Using the following Boundary Conditions, $V = V_0$ at $r = a \text{ m}$, $V = 0$ at $r = b \text{ m}$ and $b > a$.

(or)

Derive an expression for Capacitance of Cylindrical Capacitor using Laplace equation

(or)

Using Laplace equation prove that the potential distribution at any point in the region between two Co-centric cylinders of radii A & B as

$$V = V_0 \frac{\ln(r/B)}{\ln(A/B)}$$

(or)

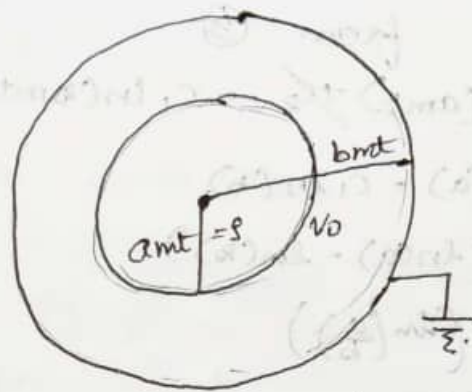
Use Laplace equation to find the Capacitance per unit length of Co-axial cable of inner radius, ~~lower~~ ~~case~~ $a \text{ m}$ and outer radius $b \text{ m}$, assume $V = V_0$ at $R = a$ and $V = 0$ at $R = b$

(or)

Apply Laplace equation show that the potential in space between two conductors of Co-axial cable of infinite length is

$$V = V_0 \frac{\ln(r/R_2)}{\ln(R_1/R_2)}$$

Ans!



$$\nabla^2 V = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0 \rightarrow (A)$$

$$\frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\int \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = \int 0$$

$$s \frac{\partial V}{\partial s} = C_1$$

$$\frac{\partial V}{\partial s} = \frac{C_1}{s} \rightarrow (1)$$

$$V = C_1 \ln(s) + C_2 \rightarrow (2)$$

Since V is a function of s , the second and third terms in the above expression (A) is considered 0.

Integrate above expression (A) with respect to s thus eqn (1) obtain.

Integrate eqn (1) again w.r.t 's'.

At $s = amt$ $V = V_0$.

$$\therefore V_0 = C_1 \ln(amt) + C_2 \rightarrow (3)$$

at $s = bmt$ $V = 0$

$$0 = C_1 \ln(bmt) + C_2 \rightarrow (4)$$

Subtract ④ from ③.

$$\begin{aligned} V_0 - 0 &= C_1 \ln(a/b) + \cancel{C_2} - C_1 \ln(b/a) - \cancel{C_2} \\ &= C_1 \ln(a) - C_1 \ln(b) \\ &= C_1 (\ln(a) - \ln(b)) \\ &= C_1 \left(\ln\left(\frac{a}{b}\right) \right) \end{aligned}$$

$$\boxed{V_0 = C_1 \ln\left(\frac{a}{b}\right)} \longrightarrow (5)$$

$$C_1 = \frac{V_0}{\ln(a/b)} \longrightarrow (6)$$

from eq ④

$$C_2 = -C_1 \ln(b)$$

$$C_2 = -\left(\frac{V_0}{\ln(a/b)} \ln(b)\right) \longrightarrow (7)$$

eqn ② $\Rightarrow$

$$V = -C_1 \ln(b) \ln(r) + \frac{V_0}{\ln(a/b)} \ln(b).$$

$$V = +\left(\frac{V_0}{\ln(a/b)} \ln(r) - \frac{V_0}{\ln(a/b)} \ln(b)\right)$$

$$= \frac{V_0}{\ln(a/b)} [\ln(r) - \ln(b)]$$

$$V = \frac{V_0}{\ln(a/b)} \ln\left(\frac{r}{b}\right) \longrightarrow (8)$$

$$E = -\nabla V = \frac{Q}{A} = \rho_s$$

$$E = -\frac{\partial V}{\partial s} = \frac{Q}{A} = \rho_c$$

$$E = -\frac{C_1}{r} = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$Q = -\frac{C_1 A}{S} \rightarrow (9)$$

The area of co-axial cable of radius S is given by,

$$A = 2\pi SL$$

where $L \rightarrow$ length of co-axial cable.

$\therefore$ The equation (9),

$$Q = -\frac{V_0}{\ln(a/b)} (2\pi SL)$$

$$Q = -\frac{V_0}{\ln(a/b)} (2\pi L)$$

$$\boxed{Q = \frac{V_0 2\pi L}{\ln(b/a)}} \rightarrow (10)$$

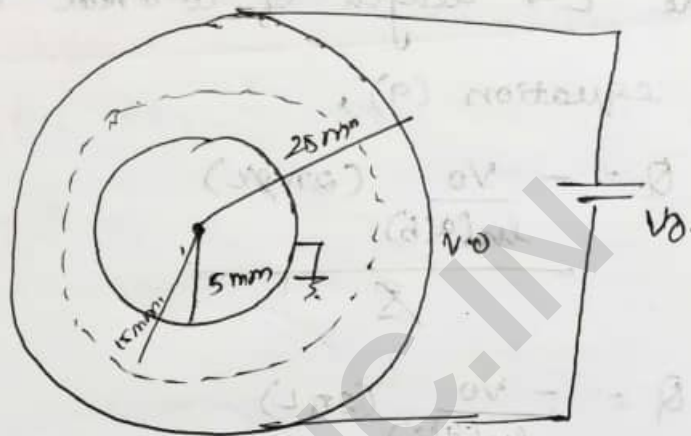
$$C = \frac{Q}{V_0}$$

$$= \frac{V_0 2\pi L}{\ln(b/a)} \cdot \frac{1}{V_0}$$

$$\boxed{C = \frac{2\pi L}{\ln(b/a)}} //$$

1. long concentric and right conducting sphere in free space $R=5\text{mm}$ and at $R=25\text{mm}$ in cylindrical co-ordinates have voltages of zero and V_0 respectively. If the E intensity $E = 8.28 \times 10^3 \text{ V/m}$ at $r=15\text{mm}$. Starting from Laplace equation. Find V_0 and charge density as the Outer Conductance. Take $\epsilon = 8.854 \times 10^{-12} \text{ F/m}$.

81



$$\nabla^2 V = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\frac{\partial V}{\partial s} = \frac{C_1}{s} \longrightarrow (1)$$

$$V = C_1 \ln(s) + C_2 \longrightarrow (2)$$

at $s=5\text{mm}$, $V=0$.

$$(2) \Rightarrow 0 = C_1 \ln(5\text{mm}) + C_2$$

$$C_2 = -C_1 \ln(5\text{mm})$$

at $s=25\text{mm}$, $V=V_0$

$$V_0 = C_1 \ln(25\text{mm}) + C_2$$

$$V_0 = C_1 \ln(25\text{mm}) - C_1 \ln(5\text{mm})$$

$$= C_1 (\ln(25\text{mm}) - \ln(5\text{mm}))$$

$$V_0 = C_1 \ln\left(\frac{25}{r}\right)$$

$$V_0 = C_1 \ln(5)$$

$$C_1 = \frac{V_0}{\ln(5)}$$

$$C_2 = -\frac{V_0}{\ln(5)}$$

$$C_2 = -V_0$$

$$C_2 = -\frac{V_0}{\ln(5)} \ln(5 \text{ mm})$$

$$V = \frac{V_0}{\ln(5)} \ln(5) + \frac{V_0}{\ln(5)} \ln(5 \text{ mm})$$

$$= \frac{V_0}{\ln(5)} \left[\ln(5) - \ln(5 \text{ mm}) \right]$$

$$V = \frac{V_0}{\ln(5)} \left[\ln\left(\frac{5}{5 \text{ mm}}\right) \right]$$

$$E = -\nabla V = \frac{V_0}{s}$$

$$E = -\frac{\partial V}{\partial s} = \frac{Q}{A} = \frac{V_0}{s}$$

$$E = -\frac{\partial V}{\partial s} = -\frac{C_1}{s} = \frac{V_0}{s}$$

$$\text{at } s = 15, V = ?$$

$$E = -\frac{C_1}{s}$$

$$E = -\frac{V_0}{s \ln(5)}$$

$$V_0 = -E s \ln(5) a_s$$

$$V_0 = -8.28 \times 10^3 \frac{\text{V}}{\text{m}} \times 15 \times 10^{-3} \text{ m} \times \ln(5) = -199.89 \text{ V}$$

$$D = \epsilon_0 E = \frac{Q}{A} = S_s$$

$$S_s = \epsilon_0 \times E$$

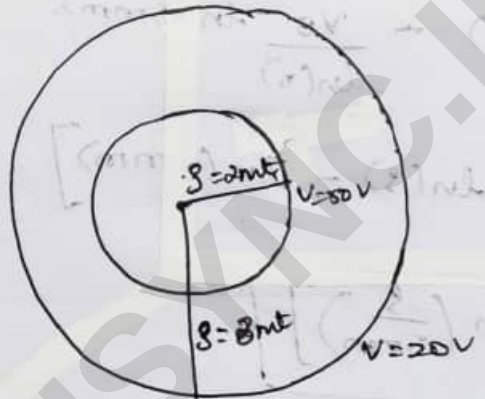
$$S_s = 8.854 \times 10^{-12} \times \left(\frac{-V_0}{s \ln(\epsilon)} \right) a_s =$$

$$S_s = 8.854 \times 10^{-12} \left(\frac{-(-199.89) V}{25 \text{ mm } \ln(\epsilon)} \right) \quad (1)$$

$$S_s = 43.98 \text{ nC/m}$$

2. Find E at $P(3, 1, 2)$ for the field of
i) two coaxial conducting cylinders. $V = 50 \text{ V}$ at
 $s = 2 \text{ mtr}$ and $V = 20 \text{ V}$ at $s = 3 \text{ mtr}$

8d



$$\nabla^2 V = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\frac{\partial V}{\partial s} = \frac{C_1}{s}$$

$$V = C_1 \ln(s) + C_2$$

at $s = 2 \text{ mtr}$, $V = 50 \text{ V}$

$$50 = C_1 \ln(2) + C_2 \rightarrow (1)$$

at $x = 3 \text{ m}$, $V = 20 \text{ V}$

$$20 = C_1 \ln(3) + C_2 \rightarrow (2)$$

Solving (1) & (2).

$$50 = C_1 \ln(2) + C_2$$

$$20 = C_1 \ln(3) + C_2$$

$$\begin{array}{r} \text{---} \\ \text{---} \end{array}$$

$$30 = -0.405 C_1$$

$$C_1 = \frac{30}{-0.405}$$

$$C_1 = -74.07 \text{ F}$$

$$20 = (-74.07) \ln(3) + C_2$$

$$C_2 = 20 + 74.07 \ln(3)$$

$$C_2 = 101.37 \text{ F}$$

$$V = C_1 \ln(x) + C_2$$

$$V = -74.07 \ln(x) + 101.37$$

$$V = 19.99 \text{ V} \approx 20 \text{ V}$$

$$E = -\frac{\partial V}{\partial x}$$

$$E = -\frac{C_1}{x}$$

$$E = -\frac{(-74.07)}{3}$$

$$E = 24.69 \text{ V/m}$$

$P(3, 1, 2)$ is given in (x, y, z) format, to convert into (ρ, ϕ, z) format.

$$\rho = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$z = z$$

$$\rho = \sqrt{3^2 + 1^2}$$

$$\boxed{\rho = \sqrt{10}}$$

$$\phi = \tan^{-1}\left(\frac{1}{3}\right)$$

$$\boxed{\phi = 0.321}$$

$$\boxed{z = 2 \text{ mtr}}$$

$$E = \frac{73.989}{\sqrt{10}} = 23.397 \text{ V/mtr}$$

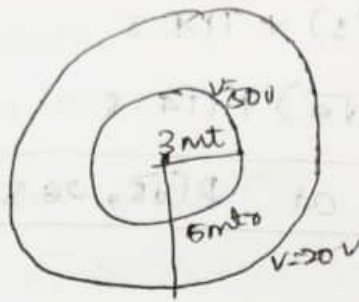
$$\boxed{E = 23.39 \text{ V/mtr}}$$

3. Assume $V = V_0$ at $\rho = a$ and $V = 0$ at $\rho = b$ in cylindrical co-ordinate $V = 865 \text{ V}$ at $\rho = 50 \text{ mtr}$. and $E = 748.2$ at $\rho = 85 \text{ mtr}$. Determine the location of voltage reference if the potential depends only on ρ .

4. find V at $(2, 1, 3)$ for the field of
 i) Two concentric conducting spheres with $V = 50 \text{ V}$ at $\rho = 3 \text{ mtr}$. and $V = 20 \text{ V}$ at $\rho = 5 \text{ mtr}$.

9) Ans

V at (2, 1, 3)



$$\nabla^2 V = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\frac{\partial V}{\partial s} = \frac{C_1}{s}$$

$$V = C_1 \ln(s) + C_2$$

at $s = 3\text{m}$, $V = 50\text{V}$

$$50 = C_1 \ln(3) + C_2 \rightarrow (1)$$

at $s = 5\text{m}$, $V = 20\text{V}$

$$20 = C_1 \ln(5) + C_2 \rightarrow (2)$$

$$50 - 20 = C_1 \ln(3) + \cancel{C_2} - C_1 \ln(5) - \cancel{C_2}$$

$$30 = C_1 (\ln(3) - \ln(5))$$

$$30 = C_1 \ln\left(\frac{3}{5}\right)$$

$$C_1 = \frac{30}{\ln(3/5)}$$

$$C_1 = -58.72 \text{ F}$$

$$C_2 = 20 - C_1 \ln(5)$$

$$= 20 + 58.72 \ln(5)$$

$$C_2 = 114.5 \text{ F}$$

$$V_1 = C_1 \ln(s) + C_2$$

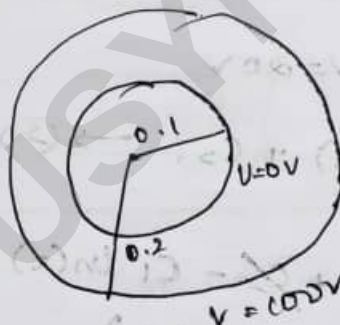
$$= -58.72 \ln(3) + 114.8$$

$$= -58.72 \ln(\sqrt{5}) + 114.8$$

$$V = 67.24 \text{ V at } P(\sqrt{5}, 26.56, 3)$$

5. An assembly of 2 concentric shell is considered. The inner shell is at a distance of 0.1 mtr and potential at 0V. The outer shell is at a distance of 0.2 mtr & the potential of 100V. The medium between them is free space. Find ϵ and d using cylindrical Co-ordinate system.

Sol



$$\nabla^2 V = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) = 0$$

$$\frac{\partial V}{\partial s} = \frac{C_1}{s}$$

$$V = C_1 \ln(s) + C_2$$

at $r = 0.1 \text{ m}$, $V = 0 \text{ V}$

$$0 = C_1 \ln(0.1) + C_2$$

$$C_2 = -C_1 \ln(0.1)$$

at $r = 0.2 \text{ m}$, $V = 100 \text{ V}$

$$100 \text{ V} = C_1 \ln(0.2) + C_2$$

$$100 \text{ V} = C_1 \ln(0.2) - C_1 \ln(0.1)$$

$$100 \text{ V} = C_1 (\ln(0.2/0.1))$$

$$C_1 = \frac{100 \text{ V}}{0.693}$$

$$\boxed{C_1 = 144.3}$$

$$C_2 = -C_1 \ln(0.1)$$

$$= -144.3 \ln(0.1)$$

$$\boxed{C_2 = 332.26}$$

$$V = C_1 \ln(r) + C_2$$

$$\boxed{V = 144.3 \ln(r) + 332.26}$$

$$E = -\nabla V = -\frac{\partial V}{\partial r}$$

$$E = -\frac{C_1}{r}$$

$$E = -\frac{(144.3)}{r}$$

$$\boxed{E = -\frac{144.3}{r}}$$

$$d = \epsilon_0 E$$

$$= \epsilon_0 \left(-\frac{144.3}{r}\right)$$

$$\boxed{d = -\frac{144.3}{r} \epsilon_0}$$

• MODULE: 3(b)

STEADY MAGNETIC FIELD.

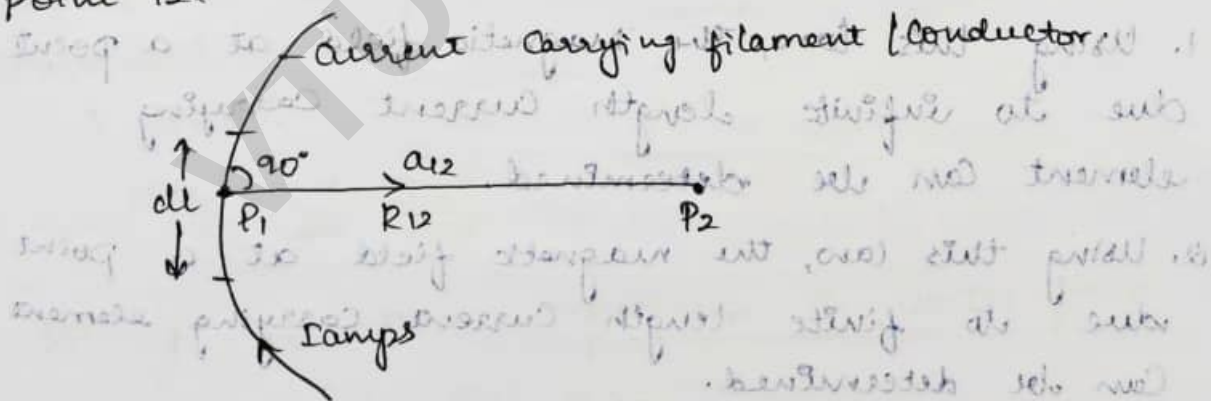
• Biot Savart Law:

→ Sources of magnetic field [H]

1. Permanent magnet
2. Changing electric field with time.
3. It is DC Current.

• Definition:

The magnitude of differential magnetic field intensity at a point P_2 is proportional to the product of current I and differential element length dl and $\sin$ of angle between current carrying element and the line joining the point P_2 to the element and is inversely proportional to square of the distance between the current carrying element and the point P_2 .



$$dH \propto \frac{I dl \sin \theta}{(R_{12})^2}$$

$$dH = \frac{1}{4\pi} \frac{I dl \sin \theta}{(R_{12})^2} A/m$$

$$dH_2 = \frac{1}{4\pi} \cdot \frac{Idl}{(R_{12})^2} \quad A/m \quad [\because \sin 90^\circ = 1] \quad \text{[Normal form]}$$

$$dH_2 = \frac{1}{4\pi} \frac{Idl}{(R_{12})^2} \times a_{12}$$

$$= \frac{1}{4\pi} \frac{Idl}{(R_{12})^2} \times \frac{R_{12}}{|R_{12}|}$$

$$dH_2 = \frac{1}{4\pi} \frac{Idl \times R_{12}}{(R_{12})^3} \quad A/m \quad \text{[Vector form]}$$

Integrating the above expression gives magnetic field at point P_2 due to entire current carrying element.

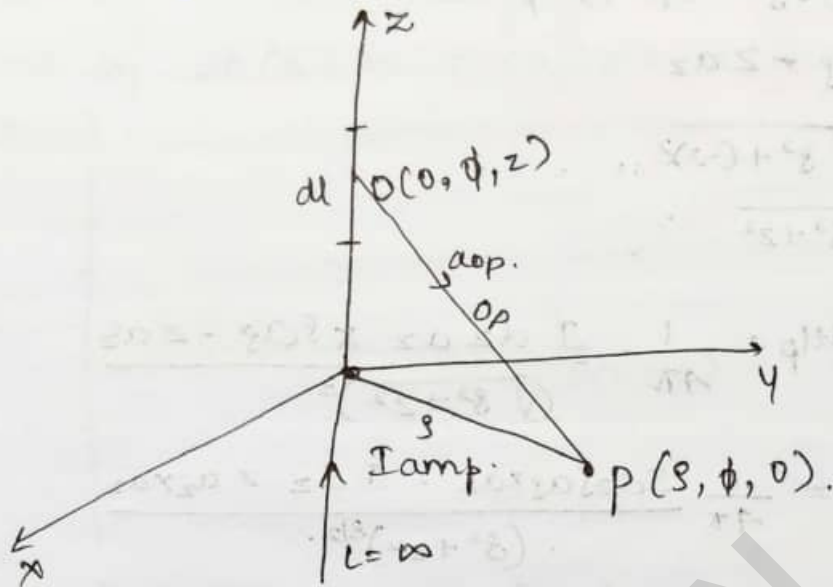
$$\int dH_2 = \int \frac{1}{4\pi} \frac{Idl}{(R_{12})^3} \times R_{12}$$

$$H_2 = \frac{1}{4\pi} \frac{Idl \times R_{12}}{(R_{12})^3} \quad A/m$$

• Application of Biot-Savart law.

1. Using this law, the magnetic field at a point due to infinite length current carrying element can be determined.
2. Using this law, the magnetic field at a point due to finite length current carrying element can be determined.
3. Due to a circular current carrying loop.
4. Due to a square current carrying loop.
5. Due to a solenoid.

1. Magnetic field at point P due to an infinite length current carrying element.



$$dH_P \propto \frac{Idl \sin \theta}{(OP)^2}$$

$$dH_P = \frac{1}{4\pi} \frac{Idl \sin \theta}{(OP)^2}$$

Since, the current carrying element is normal to the plane where the point P is considered.

$$\therefore \theta = 90^\circ$$

$$dH_P = \frac{1}{4\pi} \frac{Idl}{(OP)^2}$$

Writing the above expression in vector form

$$dH_P = \frac{1}{4\pi} \frac{Idl}{(OP)^2} \times a_{op}$$

$$dH_P = \frac{1}{4\pi} \frac{Idl \times OP}{(OP)^3} \rightarrow (A)$$

From the figure,

$$dl = dz$$

$$Idl = Idz$$

$$Idl = Idz \hat{a}_z, \rightarrow (1)$$

$$OP = (s-0)\hat{a}_s + (\phi-\phi)\hat{a}_\phi + (0-z)\hat{a}_z$$

$$OP = s\hat{a}_s - z\hat{a}_z$$

$$|OP| = \sqrt{s^2 + (-z)^2}$$

$$|OP| = \sqrt{s^2 + z^2}$$

$$\therefore \textcircled{A} \Rightarrow dH_P = \frac{1}{4\pi} \frac{I dz \hat{a}_z \times s\hat{a}_s - z\hat{a}_z}{(\sqrt{s^2 + z^2})^3}$$

$$= \frac{1}{4\pi} \frac{I dz s\hat{a}_s \times \hat{a}_z - I dz z \hat{a}_z \times \hat{a}_z}{(s^2 + z^2)^{3/2}}$$

$$[\hat{a}_z \times \hat{a}_z = 0]$$

$$= \frac{1}{4\pi} \frac{I dz s \hat{a}_\phi - I dz \times 0}{(s^2 + z^2)^{3/2}}$$

$$dH_P = \frac{1}{4\pi} \frac{I dz s \hat{a}_\phi}{(s^2 + z^2)^{3/2}}$$

Integrating the above expr, results in magnetic field at Point P due to infinite length current carrying element.

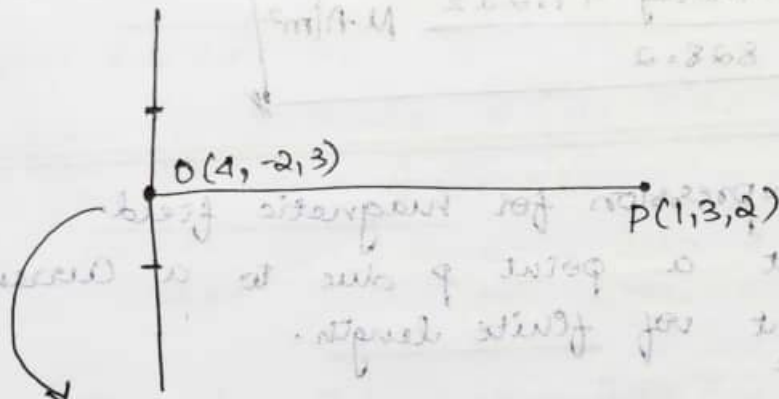
$$\int dH_P = \int_{-\infty}^{\infty} \frac{1}{4\pi} \frac{I dz s \hat{a}_\phi}{(s^2 + z^2)^{3/2}}$$

$$\boxed{H_P = \frac{I}{2\pi s} \hat{a}_\phi \text{ A/m}}$$

• Problems

1. Find the magnetic field strength at point $P(1, 3, 2)$ caused by current carrying element is defined by $2\pi(0.6a_x - 0.8a_y)\mu A\cdot m$ situated at $(4, -2, 3)$

Sol



$$Idl = 2\pi(0.6a_x - 0.8a_y)\mu A\cdot m$$

$$OP = (1-4)a_x + (3+2)a_y + (2-3)a_z$$

$$OP = -3a_x + 5a_y - 1a_z$$

$$|OP| = \sqrt{(-3)^2 + 5^2 + (-1)^2}$$

$$|OP| = \sqrt{9 + 25 + 1}$$

$$|OP| = \sqrt{35}$$

$$dH_P = \frac{1}{4\pi} \frac{Idl \times OP}{(OP)^3}$$

$$= \frac{1}{4\pi} \frac{(2\pi(0.6a_x - 0.8a_y)) \times (-3a_x + 5a_y - a_z)}{(\sqrt{35})^3}$$

$$dH_P = \frac{1}{4\pi} \frac{(1.2\pi a_x - 1.6\pi a_y) \times (-3a_x + 5a_y - a_z)}{(\sqrt{35})^3}$$

$$Idl \times OP = \begin{bmatrix} a_x & a_y & a_z \\ 1.2\pi & -1.6\pi & 0 \\ -3 & 5 & -1 \end{bmatrix}$$

$$= a_x [1.6\pi] - a_y [-1.2\pi] + a_z [6\pi - 4.8\pi]$$

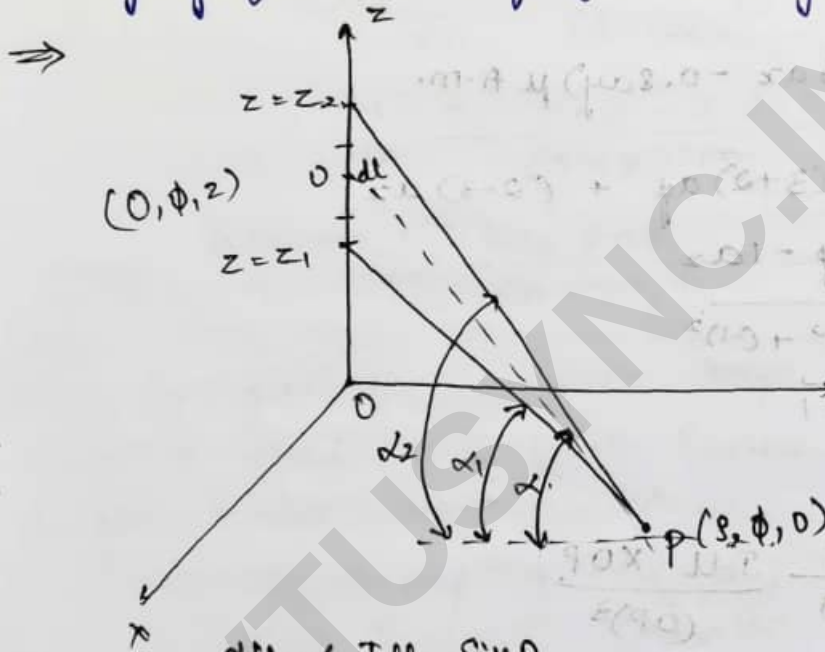
$$Idl \times OP = 1.6\pi a_x + 1.2\pi a_y + 1.2\pi a_z \mu A/m^2$$

$$dH_p = \frac{1}{4\pi} \frac{1.6\pi ax + 1.2\pi ay + 1.2\pi az}{(\sqrt{35})^3}$$

$$= \frac{1}{828.2\pi} (1.6ax + 1.2ay + 1.2az)$$

$$dH_p = \frac{1.6ax + 1.2ay + 1.2az}{828.2} \text{ M-A/m}^2$$

2. Derive an expression for magnetic field Intensity (H) at a point p due to a Current Carrying filament of finite length.



$$dH_p \propto \frac{Idl \sin\theta}{(OP)^2}$$

$$dH_p = \frac{1}{4\pi} \frac{Idl \sin\theta}{(OP)^2}$$

$$\theta = 90^\circ$$

$$dH_p = \frac{1}{4\pi} \frac{Idl}{(OP)^2}$$

in Vector form,

$$dH_p = \frac{1}{4\pi} \frac{Idl \times \vec{OP}}{(OP)^3}$$

$$dH_p = \frac{1}{4\pi} \frac{Idl \times \vec{OP}}{(OP)^3}$$

from figure

$$dl = dz$$

$$Idl = Idz$$

$$Idl = Idz \text{ as}$$

$$OP = \sqrt{s^2 + z^2}$$

$$|OP| = \sqrt{s^2 + z^2}$$

$$d\vec{H} = \frac{1}{4\pi} \frac{Idz \vec{a}_z \times \vec{r}}{(\sqrt{s^2 + z^2})^3}$$

$$= \frac{1}{4\pi} \frac{Idz s \vec{a}_s \times \vec{a}_z - Idz z \vec{a}_z \times \vec{a}_z}{(s^2 + z^2)^{3/2}}$$

$$= \frac{1}{4\pi} \frac{Idz s \vec{a}_\phi - Idz \times 0}{(s^2 + z^2)^{3/2}}$$

$$d\vec{H} = \frac{1}{4\pi} \frac{Idz s \vec{a}_\phi}{(s^2 + z^2)^{3/2}}$$

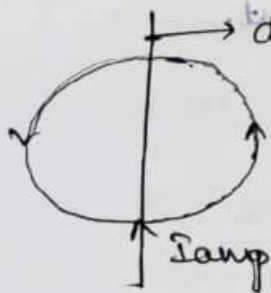
On Integrating

$$\int d\vec{H} = \int_{z_1}^{z_2} \frac{1}{4\pi} \frac{Idz s \vec{a}_\phi}{(s^2 + z^2)^{3/2}}$$

$$\vec{H} = \frac{I}{4\pi s} [\sin(\alpha_2) - \sin(\alpha_1)] \vec{a}_\phi \text{ A/m}$$

• Ampere Circuit Law.

> Statement: The line integral of magnetic field intensity H around a single closed path is equal to the current enclosed by that path.



The magnetic field is circular in nature.

$$\oint H \cdot dl = I_{amp}$$

• Proof:

consider a current carrying conductor, placed along z-axis

The magnetic field (H) around the conductor is represented by circle.

The magnetic field due to an infinite length conductor is given by,

$$H = \frac{I}{2\pi r} \text{ A/m.} \quad \longrightarrow (1)$$

Multiply the equation (1) by dl on B.S

$$H \cdot dl = \frac{I}{2\pi r} \text{ A} \cdot \text{m}$$

Since $dl = r d\phi$ A/m,

$$H \cdot dl = \frac{I}{2\pi r} \cdot r d\phi \text{ A} \cdot \text{m}$$

$$H \cdot dl = \frac{I}{2\pi} d\phi \quad \longrightarrow (2)$$

Integrate eq (2) w.r.t $d\phi$.

$$\int H \cdot dl = \int_0^{2\pi} \frac{I}{2\pi} d\phi$$

$$\int H \cdot dl = \frac{I}{2\pi} [2\pi - 0]$$

$$\boxed{\int H \cdot dl = I}$$

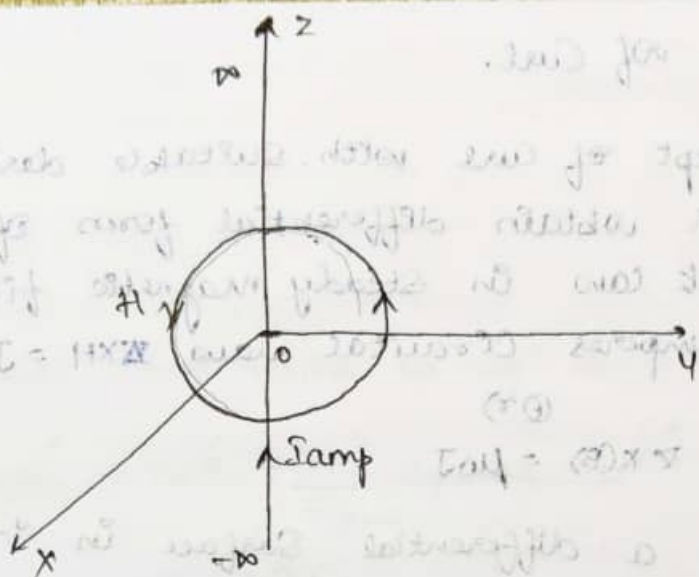
→ Application of ampere circuit law.

**

1. Derive an expression for magnetic field intensity H due to infinitely long straight current carrying conductor / filament.



⇒



- The magnitude of H depends on radius S
- The direction of the magnetic field is always tangential to a closed path.
- The magnetic field has only one component in the ϕ direction i.e. H_ϕ .

Since H has only one component in the ϕ direction.

$$H = H_\phi \text{ a}_\phi \text{ A/m} \longrightarrow (A).$$

The length of differential element is given by

$$dl = S d\phi \text{ a}_\phi$$

Multiply eq(A) By dl on B.S.

$$H \cdot dl = H_\phi \text{ a}_\phi \cdot S d\phi \text{ a}_\phi$$

$$H \cdot dl = H_\phi S d\phi$$

$$H \cdot dl = H_\phi S d\phi$$

(4) Using Ampere Circuit law,

Integrate on B.S.

$$I = \int H \cdot dl = \int_0^{2\pi} H_\phi S d\phi$$

$$= S H_\phi \int_0^{2\pi} d\phi$$

(Consider H_ϕ as constant as it is the magnetic field component)

$$I = S H_\phi (2\pi)$$

$$H_\phi = \frac{I}{2\pi S} \text{ A/m}$$

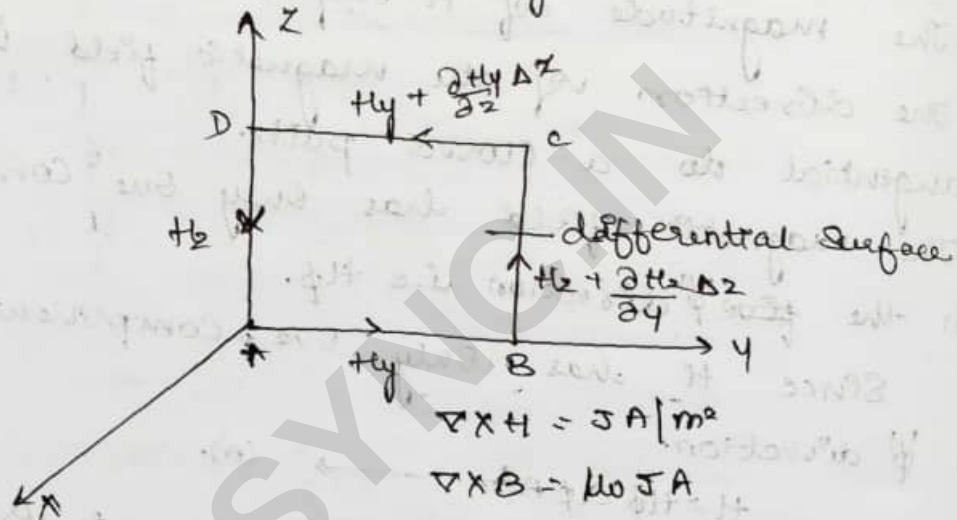
The Concept of Curl.

1. Explain concept of curl with suitable derivation for curl H or obtain differential form of ampere's law in steady magnetic field. or prove that ampere's circuital law $\nabla \times H = J \text{ A/m}^2$

(Q.7)

show that $\nabla \times B = \mu_0 J$

$\Rightarrow$ Consider a differential surface in y - z plane in Cartesian co-ordinate system.



Using the curl, the total current I , crossing the surface is measured.

According to ampere circuit law,

$$\oint_{\text{loop}} H \cdot dl = I$$

$$I = \int_A^B H_y \cdot dl + \int_B^C \left(H_z + \frac{\partial H_z}{\partial y} \Delta y \right) \cdot dl + \int_C^D \left(H_y + \frac{\partial H_y}{\partial z} \Delta z \right) \cdot dl + \int_D^A H_z \cdot dl \quad \text{--- (A)}$$

$$\int_A^B H_y \cdot dl = \int_A^B H_y \Delta y = H_y \Delta y$$

$$\int_B^C \left(H_z + \frac{\partial H_z}{\partial y} \Delta y \right) \Delta z = H_z \Delta z + \frac{\partial H_z}{\partial y} \Delta y \cdot \Delta z$$

$$\int_C H_y + \frac{\partial H_y}{\partial z} \Delta z (-\Delta y) = -H_y \Delta y - \frac{\partial H_y}{\partial z} \Delta z \Delta y$$

$$\int_D H_z (-\Delta z) = -H_z \Delta z$$

Substitute all terms in eqn (A)

$$I = H_y \Delta y + H_z \Delta z + \frac{\partial H_z}{\partial y} \Delta y \cdot \Delta z - H_y \Delta y - \frac{\partial H_y}{\partial z} \Delta z \Delta y - H_z \Delta z$$

$$I = \frac{\partial H_z}{\partial y} \Delta y \cdot \Delta z - \frac{\partial H_y}{\partial z} \Delta z \Delta y$$

$$\int H \cdot dl = \Delta z \Delta y \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) = J \Delta y \Delta z$$

$$\boxed{\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = J_z}$$

$$\frac{\partial H_z}{\partial z} - \frac{\partial H_x}{\partial x} = J_x$$

$$\boxed{\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = J_x}$$

$$\boxed{J_y = \frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z}}$$

Note

$$J = \frac{I}{A} = \frac{I}{\Delta y \Delta z}$$

$$I = J \Delta y \Delta z$$

Since in Cartesian Co-ordinate System, J is given by,

$$J = J_x a_x + J_y a_y + J_z a_z$$

$$J = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) a_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) a_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) a_z$$

$$\boxed{J = \nabla \times H}$$

Since, $B = \mu_0 H$

$$\therefore H = \frac{B}{\mu_0}$$

$$\boxed{\mu_0 J = \nabla \times B}$$

PROBLEMS

1. In Cylindrical co-ordinates, the magnetic field is given as $H = (4s - 2s^2)a_\phi$ A/m. For $0 \leq s \leq 1$ m. Find

i) Current density

ii) Total Current that passes through the surface $z=0$, $0 \leq s \leq 1$ m in a_z direction.

Ans $P(s, \phi, z)$

$$\underline{J = \nabla \times H.}$$

$$\nabla \times H = \frac{1}{s} \begin{bmatrix} a_s & s a_\phi & a_z \\ \frac{\partial}{\partial s} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ H_s & s H_\phi & z \end{bmatrix} \rightarrow \textcircled{1}$$

$$\nabla \times H = H_s a_s + H_\phi a_\phi + H_z a_z$$

$$\textcircled{1} \rightarrow = \frac{1}{s} \begin{bmatrix} a_s & s a_\phi & a_z \\ \frac{\partial}{\partial s} & 0 & 0 \\ 0 & s(4s - 2s^2) & 0 \end{bmatrix}$$

$$= \frac{1}{s} [0 - s a_\phi [0 - 0] + a_z \left[\frac{\partial}{\partial s} s(4s - 2s^2) \right]]$$

$$= \frac{1}{s} \left(\frac{\partial}{\partial s} (4s^2 - 2s^3) \right) a_z$$

$$= \frac{1}{s} [8s - 6s^2] a_z$$

$$\boxed{\nabla \times H = [8 - 6s] a_z}$$

$$I = \int J \cdot ds$$

$$= \int (8 - 6s) a_z \cdot s d\phi ds a_z$$

$$= \int_0^1 (8s - 6s^2) ds \int_0^{2\pi} d\phi = \left[\frac{8s^2}{2} - \frac{6s^3}{3} \right]_0^1 [2\pi]$$

$$= (4 - 2)(2\pi)$$

$$= 2(2\pi)$$

$$\boxed{I = 4\pi \text{ A}}$$

2. For a given Current density $J = 10^3 \sin \theta \text{ ar A/m}^2$ in Spherical Co-ordinate System. Find the current passing the spherical shell with the radius $r = 0.02 \text{ m}$.

Ans $J = 10^3 \sin \theta \text{ ar A/m}^2$

1) $I = ?$ $r = 0.02 \text{ m}$

$$I = \int J \cdot d\mathbf{s}$$

$$= \int 10^3 \sin \theta \hat{r} \cdot r^2 \sin \theta d\theta d\phi \hat{r}$$

$$= \int 10^3 \sin^2 \theta d\theta d\phi$$

$$= 10^3 \int_0^\pi \sin^2 \theta d\theta \int_0^{2\pi} d\phi \cdot r^2$$

$$= 10^3 \int_0^\pi \frac{1 - \cos 2\theta}{2} d\theta \int_0^{2\pi} d\phi \times 0.02^2$$

$$\boxed{I = 3.94 \text{ amp}}$$

• STOKES'S THEOREM

Integration of any vector around the closed path is always equal to integration of the curve of that vector throughout the surface enclosed.

$$\therefore \oint_C \mathbf{H} \cdot d\mathbf{l} = \oint_S \nabla \times \mathbf{H} \cdot d\mathbf{s}$$

• Proof:

From ampere circuital law,

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{amps}} \longrightarrow (1)$$

From the concept of Current density (J),

$$I = \oint_S J \cdot d\mathbf{s} \longrightarrow (2)$$

equating eq (1) and (2),

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \oint_S \mathbf{J} \cdot d\mathbf{s} \longrightarrow (3)$$

From ampere's circuital law,

$$\nabla \times \mathbf{H} = \mathbf{J} \text{ A/m.} \longrightarrow (4)$$

Substitute for $\mathbf{J}$ in (3)

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \oint_S \nabla \times \mathbf{H} \cdot d\mathbf{s} \longrightarrow (5)$$

• PROBLEMS:

1. Verify Stokes theorem for a given field, $\mathbf{H} = 6xy \mathbf{a}_x - 3y^2 \mathbf{a}_y \text{ A/m}$, and a rectangular path around the region, defined by, $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $z = 0$.

Ans

$$\nabla \times \mathbf{H} = \begin{bmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{bmatrix}$$

$$\nabla \times \mathbf{H} = \begin{bmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & 0 \\ 6xy & (-3y^2) & 0 \end{bmatrix}$$

$$= \mathbf{a}_x [0] - \mathbf{a}_y [0] + \mathbf{a}_z \left[-\frac{\partial}{\partial x} 3y^2 - \frac{\partial}{\partial y} 6xy \right]$$

$$= -0 - 6x \mathbf{a}_z$$

$$\boxed{\nabla \times \mathbf{H} = -6x \mathbf{a}_z}$$

$$\oint_S \nabla \times \mathbf{H} \cdot d\mathbf{s} = \oint_S -6x \mathbf{a}_z \cdot d\mathbf{s}$$

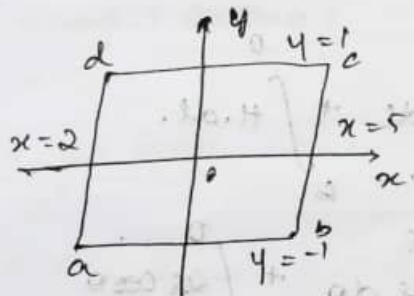
$$\oint_S \nabla \times \mathbf{H} \cdot d\mathbf{s} = \oint_S -6xz \mathbf{a}_z \cdot d\mathbf{y} dz \mathbf{a}_z.$$

$$= -6 \int_2^5 dz \int_{-1}^1 dy$$

$$= -\frac{6}{2} [25 - 4] [1 + 1]$$

$$= -\frac{6}{2} (21)(2)$$

$$\boxed{\oint_S \nabla \times \mathbf{H} = -12 \text{ A}} \quad \text{amps}$$



$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_a^b \mathbf{H} \cdot d\mathbf{l} + \int_b^c \mathbf{H} \cdot d\mathbf{l} + \int_c^d \mathbf{H} \cdot d\mathbf{l} + \int_d^a \mathbf{H} \cdot d\mathbf{l}$$

$$= \int_a^b 6xy \, dx + \int_b^c -3y^2 \, dy + \int_c^d 6xy \, (-dx) + \int_d^a -3y^2 \, (-dy)$$

$$= \int_2^5 6xy \, dx - 3 \int_{-1}^1 y^2 \, dy - \int_5^2 6xy \, dx + \int_1^{-1} 3y^2 \, dy$$

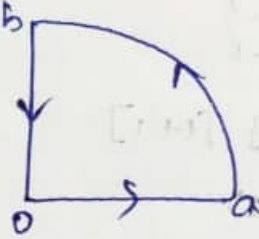
$$= 6y \left(\frac{x^2}{2} \right)_2^5 - 3 \left(\frac{y^3}{3} \right)_{-1}^1 - 6y \left(\frac{x^2}{2} \right)_5^2 + 3 \left(\frac{y^3}{3} \right)_1^{-1}$$

$$= 3y [25 - 4] - \frac{3}{3} [1 + 1] - 3y [4 - 25] + \frac{3}{3} [-1 - 1]$$

$$= 63y - 2 + 63y - 2$$

$$= 126y - 4$$

2. Verify the Stokes theorem for the field $H = 2x \cos \theta \, a_r + x a_\theta \, \text{A/m}$ for the path shown below, with $x=0$ to 1 and $\theta=0$ to 90° . Verify the Stokes theorem.



Ans

$$\oint_C H \cdot dl = \int_S \nabla \times H \cdot d\mathbf{s}$$

$$\oint_C H \cdot dl = \int_0^1 H \cdot dl + \int_a^b H \cdot dl + \int_b^0 H \cdot dl$$

$$= \int_0^1 2x \cos \theta \, dx + \int_a^b x \, d\theta + \int_b^0 2x \cos \theta \, dx$$

$$= \int_0^1 2x \cos \theta \, dx + \int_0^{\pi/2} x \, d\theta + \int_1^0 2x \cos \theta \, dx$$

$$= \cos \theta \int_0^1 2x \, dx + x \int_0^{\pi/2} d\theta + \cos \theta \int_1^0 2x \, dx$$

$$= \cos \theta \left[x^2 \right]_0^1 + x \left[\theta \right]_0^{\pi/2} + \cos \theta \left[x^2 \right]_1^0$$

$$= \cos \theta + \frac{\pi}{2} x - \cos \theta$$

$$= \cos(0) + \frac{\pi}{2}(1) + \cos 90^\circ$$

$$= 1 + \frac{\pi}{2} + 0$$

$$\boxed{\oint_C H \cdot dl = 2.57 \text{ amps}}$$

$$\nabla \times H = \begin{vmatrix} a_r & r a_\theta & r \sin \theta a_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ H_r & r H_\theta & r \sin \theta H_\phi \end{vmatrix} = \frac{1}{r^2 \sin \theta}$$

$$\begin{aligned}
 &= a_\phi \left[\frac{\partial}{\partial \theta} (r \sin \theta H_\phi) \right] - r a_\theta \left[\frac{\partial}{\partial r} (r \sin \theta H_\phi) \right] + r \sin \theta a_\phi \left[\frac{\partial}{\partial r} (r H_\theta) - \frac{\partial}{\partial \theta} (r) \right] \\
 &= a_\phi \left[\frac{\partial}{\partial \theta} (r \sin \theta \cdot 2r \cos \theta) \right] - r a_\theta \left[\frac{\partial}{\partial r} (r \sin \theta (0)) \right] \times \frac{1}{r^2 \sin \theta} \\
 &\quad + r \sin \theta a_\phi \left[\frac{\partial}{\partial r} (r(r)) - \frac{\partial}{\partial \theta} (r \cos \theta) \right] \times \frac{1}{r^2 \sin \theta} \\
 &= r \sin \theta \left\{ a_\phi [2r] - 2r(-\sin \theta) \right\} \times \frac{1}{r^2 \sin \theta} \\
 &= r \sin \theta (2r a_\phi + 2r \sin \theta) \times \frac{1}{r^2 \sin \theta} \\
 &= \frac{r \sin \theta (2r a_\phi + 2r \sin \theta)}{r^2 \sin \theta} \\
 &= \frac{2r a_\phi + 2r \sin \theta}{r} \\
 &= 2a_\phi + 2\sin \theta
 \end{aligned}$$

3 For given $H = 20r^2 a_\phi$ A/m. Determine the J also determine the total current that cross the surface $r = 1$ mtr $0 < \phi < 2\pi$, $z = 0$ in cylindrical co-ordinate system.

Ans $J = \nabla \times H$

$$\nabla \times H = \frac{1}{r} \begin{bmatrix} a_r & r a_\phi & a_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ r & r a_\phi & H_z \end{bmatrix}$$

$$= \frac{1}{r} \begin{bmatrix} a_s & r a_\phi & a_z \\ \frac{\partial}{\partial s} & 0 & 0 \\ 0 & r \frac{\partial}{\partial r} & 0 \end{bmatrix}$$

$$= \frac{1}{r} \begin{bmatrix} r a_z \\ \frac{\partial}{\partial s} r \cdot 20 r^2 \end{bmatrix}$$

$$= \frac{1}{r} [r a_z + 20 \times 3 r^2]$$

$$\boxed{J = 60 r^2 a_z}$$

$$I = \int J \cdot ds$$

$$= \int 60 r^2 a_z \cdot ds$$

$$= \int 60 r^2 a_z \cdot r dr d\phi a_z$$

$$= \int 60 r^3 dr d\phi$$

$$= 60 \int r^3 dr \int d\phi = 60 \left[\frac{r^4}{4} \right]_0^3 \left[\phi \right]_0^{2\pi}$$

$$= \frac{60}{4} [1] 2\pi$$

$$\boxed{I = 125.6 \text{ amp}}$$

4. Calculate the value of vector current density in cylindrical coordinates at the point $P(1.5, 90^\circ, 0.5)$ at $H = \frac{2}{r} \cos(0.2\phi) a_s + 1/m$.

Ans $J = \nabla \times H$

$$\nabla \times H = \frac{1}{r} \begin{bmatrix} a_s & r a_\phi & a_z \\ \frac{\partial}{\partial s} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ H_s & r H_\phi & H_z \end{bmatrix}$$

$$= \frac{1}{r} \begin{bmatrix} a_s & r a_\phi & a_z \\ \frac{\partial}{\partial s} & \frac{\partial}{\partial \phi} & 0 \\ \frac{2}{r} \cos(0.2\phi) & 0 & 0 \end{bmatrix}$$

$$= \frac{1}{s} \left[a_1(0) - s a_1(0) + a_2 \left(0 - \frac{\partial}{\partial \phi} \frac{2s}{s} \cos(0.2\phi) \right) \right]$$

$$= \frac{1}{s} \left(-\frac{2s}{s} (0.2) (-\sin(0.2\phi)) \right) a_z$$

$$\nabla \times H = \frac{+0.4 \sin(0.2\phi)}{s^2} a_z$$

$$P(1.5, 90^\circ, 0.5) = J = \frac{0.4 \sin 0.2(90^\circ)}{(1.5)^2} a_z$$

$$J = 0.054 \text{ A/m}^2$$

~~5. If the magnetic field intensity in the region $0 \leq r \leq 0.5$ mtr. in cylindrical co-ordinates, the current density is $J = (1.55 - 0.5r^2) a_\phi$ A/m. $J = 10^3 \sin \theta a_\phi$~~

~~6. For a given current density $J = 10^3 \sin \theta a_\phi$ A/m~~

★ SCALAR AND VECTOR MAGNETIC POTENTIAL.

In magnetic field, there are two types of potential.

- 1) Scalar Magnetic potential (V_m)
- 2) Vector Magnetic potential (A)

1. Scalar Magnetic potential (V_m)

$$\nabla \times \nabla V = 0 \rightarrow (1)$$

$$\nabla \cdot (\nabla \times A) = 0 \rightarrow (2)$$

$\therefore$ we can't scalar magnetic potential, we can write

$$\nabla \times \nabla V_m = 0 \rightarrow (3)$$

$$B = \mu H$$

$$\mu = \mu_0 \mu_r$$

where $\mu_r = 1$ for non-magnetic material,
whereas $\mu_r > 1$.

where B is the magnetic flux density. $\rightarrow (A)$

• The magnetic flux (ϕ)

The magnetic flux of crossing any surface is calculated by using the formula.

$$\phi = \oint B \cdot ds.$$

• The magnetic flux density (B)

The total magnetic lines of force crossing a unit area in a plane normal to the direction of flux.

$$\text{Therefore } B = \frac{d\phi}{ds} \text{ Tesla.}$$

• The magnetic field intensity (H)

The quantitative measure of strength or the weakness of the magnetic field is given by the magnetic field intensity H .

xx → Relation between B and H

$$B = \mu H$$

$$\text{Since } \mu = \mu_0 \mu_r, \mu_r = 1$$

$$\mu = \mu_0.$$

$$\therefore \boxed{B = \mu_0 H}$$

$\therefore$ The value of μ_0 for free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m.}$$

- Magnetic Scalar potential V_m with the magnetic field H .

AS $E = -\nabla V$

Similarly,
 $H = -\nabla V_m$

$\therefore \nabla V_m = -H$

$\nabla \times \nabla V = 0$ (from exp (1))

$\nabla \times \nabla V_m = 0$

$\nabla \times (-H) = 0$

$\therefore \nabla \times H = 0$

The above expression is true when $J=0$.

• PROBLEM.

1. A point $P(x, y, z)$, the components of A_x, A_y, A_z of Vector magnetic potential A are given by,
 $A_x = 4x + 2y + 2z$, $A_y = 5x + 6y + 3z$ & $A_z = 2x + 3y + 5z$.
 Determine the magnitude and the direction of B at point P .

Ans $A_x = 4x + 2y + 2z$

$A_y = 5x + 6y + 3z$

$A_z = 2x + 3y + 5z$

$A = A_x a_x + A_y a_y + A_z a_z$

$B = \nabla \times A = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$

$$\nabla \times \mathbf{A} = \begin{bmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4x+3y+2z & 5x+6y+3z & 2x+3y+5z \end{bmatrix}$$

$$= a_z [3-3] + a_y [2-2] + a_x [5-3]$$

$$= 0 - 0 + 2a_x$$

$$\boxed{\nabla \times \mathbf{A} = 2a_x}$$

$$\boxed{B = 2T}$$

2. A Vector Magnetic Potential in a free space is given by $\mathbf{A} = 100 \text{ s}^{0.5} a_z$ find Magnetic field intensity $\mathbf{H}$ and the Current density $\mathbf{J}$?

$$\mathbf{A} = 100 \text{ s}^{0.5} a_z \quad \mathbf{H} = ? \quad \mathbf{J} = ? \quad \mathbf{B} = \mu_0 \mathbf{H}$$

$$\text{Ans } \mathbf{J} = \nabla \times \mathbf{H} = \frac{1}{\mu_0} \begin{bmatrix} a_x & 3a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & 3A_y & A_z \end{bmatrix}$$

$$\mathbf{B} = \nabla \times \mathbf{A} = -150 \text{ s}^{0.5} a_\phi$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} = \frac{-150 \text{ s}^{0.5} a_\phi}{4\pi \times 10^{-7} \text{ H/m}}$$

$$\boxed{\mathbf{H} = -119.36 \times 10^6 \text{ s}^{0.5} a_\phi \text{ A/m}}$$

$$\mathbf{B} = \frac{1}{s} \begin{bmatrix} a_x & 3a_y & a_z \\ \frac{\partial}{\partial x} & 0 & 0 \\ 0 & 0 & 100s \end{bmatrix}$$

$$= \frac{1}{s} [a_x(0) - 3a_y(\frac{\partial}{\partial x}(100 \text{ s}^{0.5}) - 0) + a_z(0)]$$

$$= \frac{1}{s} [-3(100 \times 0.5 \text{ s}^{-0.5}) a_y]$$

$$\boxed{\mathbf{B} = -150 \text{ s}^{0.5} a_\phi}$$

$$J = \nabla \times H = \frac{1}{s} \begin{bmatrix} a_s & s a_\phi & a_z \\ \frac{\partial}{\partial s} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_s & s A_\phi & A_z \end{bmatrix}$$

$$= \frac{1}{s} \begin{bmatrix} a_s & s a_\phi & a_z \\ \frac{\partial}{\partial s} & 0 & 0 \\ 0 & (-119.36 \mu s^{0.5}) & 0 \end{bmatrix}$$

$$= \frac{1}{s} \left[a_z \left(\frac{\partial}{\partial s} (s) - 119.36 \mu s^{0.5} \right) \right]$$

$$= \frac{a_z}{s} (-119.36 \times 10^6) (1.5)^{0.5}$$

$$\boxed{J = -179.04 s^{-0.5} a_z \text{ A/m}^2}$$

3. For a given Magnetic Vector potential $A = 20 r^2 a_\phi$. Find Magnetic flux density B in Spherical co-ordinate system.

$$\underline{\text{Ans}} = \frac{1}{r^2 \sin \theta} \begin{bmatrix} a_r & r a_\theta & r \sin \theta a_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ H_r & r H_\theta & r \sin \theta H_\phi \end{bmatrix}$$

$$= \frac{1}{r^2 \sin \theta} \begin{bmatrix} a_r & r a_\theta & r \sin \theta a_\phi \\ \frac{\partial}{\partial r} & 0 & 0 \\ 0 & 0 & r \sin \theta (20 r^2) \end{bmatrix}$$

$$= \frac{1}{r^2 \sin \theta} \left[a_r(0) - r a_\theta \left[\frac{\partial}{\partial r} (r^3 \sin \theta a_\phi - 0) \right] + r \sin \theta a_\phi(0) \right]$$

$$= \frac{1}{r^2 \sin \theta} \left[r a_\theta (20 \sin \theta (3 r^2)) \right]$$

$$= \frac{1}{r^2 \sin \theta} [60 r^3 \sin \theta a_\theta] = \boxed{B = 60 r a_\theta}$$